

01 — Giant Juncao: the crop

The C4 grass crop that renders sugarcane and maize commercially obsolete — biology, yield curves, evolution through breeding.

What this instalment covers

- The biology of *Pennisetum giganteum* × *P. americanum*: C4 photosynthesis, perennial growth, a root system that tolerates saline and alkaline soils.
- Productivity in figures: KwaZulu-Natal data, the Papoea-Nieuw-Guinea record, comparisons with switchgrass, miscanthus, and sugarcane.
- The developmental history: Professor Lin Zhanxi, Fujian Agriculture and Forestry University, stabilisation as a hybrid in 2002.
- The time window: how Chinese consortia have been exporting the crop since 2010 to Papoea-Nieuw-Guinea, Fiji, Centraal-Afrikaanse Republiek and Zimbabwe.
- Breeding potential: how maize went from 4 to 11 tonnes of grain per hectare through 70 years of breeding, and what that implies for Juncao in the coming 10–20 years.
- CRISPR precision where necessary: higher dry matter fraction, lower lignin content for cellulose conversion, frost resistance for expansion into temperate zones.

This instalment is being further developed with field photographs, yield curves, and the European breeding institutes (Wageningen, INRAE, ETH Zürich, KU Leuven) that can accelerate this evolution.